

Cross-method check against the immortal-accurate anchor

$K = 3$ deterministic CRRA REE at $(\tau, \gamma) = (2, 0.1)$,
joint-limit kernel co-area nailed to $F \leq 10^{-11}$, deficit ≈ 0.282 across $G \in \{9, 13, 17, 21, 25, 31\}$

Fixed-Point Factory

June 2026

The anchor

`projects/REZN/solved.fixed_points/k3_coarea_limit/` contains six nailed fixed points along the joint central-path $h = C\sqrt{\Delta u}$ ($C = 0.45$): residual F_∞ from $6 \cdot 10^{-15}$ at $G = 9$ to $\sim 10^{-11}$ at $G = 31$, with the one-to-one revelation deficit converging to $0.282 - 0.291$. This *is* the reference value of the $K=3$ deterministic equilibrium at this (τ, γ) — the only object in the repo I trust to four digits. The cross-check here tests other methods against it on the *same* cell.

Methods and results

method (all DD where DD)	F^{own}	F_w^{strict}	slope	deficit	$ \Delta_{\text{anchor}} $	wall
anchor (joint-limit, $G=9..31$)	$\leq 10^{-11}$	—	≈ 0.32	0.282	0.000	—
M1 legacy R4 DD $G=7$ ($h=0.2..0.5$)	$9 \cdot 10^{-29}$	0.086	0.799	0.030	0.252	71 s
M2 scaled R4 DD $G=9$ ($h=0.07..0.16$)	$3 \cdot 10^{-27}$	0.136	1.151	0.006	0.276	808 s
M3 strict- $h=0$ best-iter $G=9$	0.088	0.088	0.776	0.012	0.270	2 s

The headline

All three production-side methods miss the anchor deficit by a quarter of the unit interval. Self-residual at the arithmetic limit (M1, M2 at $10^{-27} - 10^{-29}$) is no protection. Even the bare strict- $h=0$ best-iterate (M3) misses by the same amount.

Reading

- M1 / M2 are R4 Newton-nailed fixed points of their respective operators. Their deficits are near zero because at h on the order of the cube cell spacing, the kernel-band Φ is Lipschitz, the equilibrium is essentially *fully revealing* for that operator, and DD just pins that wrong limit to 29 digits.
- M3 reaches a strict best-iterate with $F = 0.088$ — the strict- $h=0$ stall, documented in Findings 9 and 17 of the monograph. Its deficit is also near zero for the same reason: the best iterate found by damped Picard at $G = 9$ is near the FR branch.

- The anchor’s deficit comes from the *joint* limit $h \downarrow 0$ with $G \uparrow$ at fixed ratio. Neither fixed- h Newton nor strict-Picard alone follows that central path.
- This is the same lesson as the h -ladder at $(\tau = 0.1, \gamma = 10)$ from earlier: the legacy h values bias the slope by a factor of two; only the trajectory in h has economic content. At $(\tau = 2, \gamma = 0.1)$ the bias is even larger because the PR is much stronger here.

What “immortal-accurate” actually means

The anchor was produced by the consistent-kernel joint-limit operator plus high precision at six grids. Two independent confirmations exist in the repo:

- `k3_hfree_morse/` ($G = 9$, no-kernel co-area): $\|F\| = 8 \cdot 10^{-15}$, deficit ≈ 0.17 (lower because that experiment is at $(\gamma = 1, \tau = 1)$, a different cell — but the operator is independent).
- `k3_strict_h0_exact/` (flint marching-squares, ~ 130 -digit): the strict- $h=0$ operator at $(\tau = 2, \gamma = 0.1)$ *stalled* at $F = 0.995$ (no improvement from cold start) — precisely the C^0 -not- C^1 obstruction proven in Finding 9. Confirms the anchor is not reachable by strict alone, only by the joint-limit path.

The anchor is therefore not flint-level digits, but it is *operator-consistent* at six independent grids and confirmed by two other operator families. That is the strongest truth-closeness the project currently owns.

Implication for any future γ -ladder

A single- h ladder is not adequate at any τ where the equilibrium is more than mildly partial. The valid product is the h -trajectory plus the strict bound, as the $v2/v3$ design enforces. The current dual-DD $v3$ run at $(\tau = 0.1)$ is right-shaped for that regime (slope wedge $O(\tau^2/\gamma)$ is small, so single-cell DD nails carry useful information) — but the deficit it reports is the same “operator-conditional” object: the only economically defensible deliverable remains the h -trend toward the strict reference.